Project Plan

Project Name: Food Nutrition Information Webpage

Group Number: 85

Team members

Student No.	Full Name	GitHub Username	Contribution (sum to 100%)
s5290101	Ethan Weissel	ethan.weissel@griffithuni.edu.au	33.3% or Equal
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s2967734	Jayden Alexander-Morpeth	jayden.alexander-morpeth@griffithuni.edu.au	33.3% or Equal

Brief Description of Contribution

Please Describe what you have accomplished in this group project.

- s5290101, Ethan Weissel
  - Accomplishments: For this assignment I completed part of the Executive Summary and Unit testing. I was also responsible for the updating of the Software Design Doc, Project plan and Gantt Chart. I also played a very minimal role in the writing of the system code.
- s5379918, Atharav Verma
  - Accomplishments: Problem background, system capabilities/overview, potential benefits, user requirements, software requirements, use case diagrams and use cases. I contributed to coding the pie chart. I also wrote the testing, executive summary and documentation for the nutrition breakdown and nutrition comparison feature. I also reviewed and improved existing test cases of other team members.
- s2967734, Jayden Alexander-Morpeth
  - Accomplishments: For this project, I have designed the software design flowchart, written up the functions, designed the detailed design flowchart, designed the structural design flowchart, justified its design, designed the visual design mockups and justified their designs. I wrote most of the system code, developed the GUI, and did the testing for the search function and nutrition range filter.

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1. Project Overview

1.1 Project Objectives

The Food Nutritional Tool has two primary objectives: firstly, to enable users to understand the nutritional values of different foods, and secondly, to facilitate comparisons between these foods. This tool aims to help users assess the healthiness of various foods, find options that meet specific nutritional criteria, and provide a platform for individuals with particular dietary requirements to identify suitable foods.

Comparing foods is also a key feature, allowing users to make more informed choices, find healthier substitutes for the foods they currently consume, and discover similar foods to those they already enjoy.

1.2 Project Stakeholders

Many stakeholders will have a vested interest in the project both internally and externally. Internally the key stakeholders will be the project team developing the software and the company officials who will receive the finished software as a product. Externally the various end users are extremely important as they will be the ones using/purchasing the tool after the project is completed. Investors also have a keen interest in the quality and success of the project. Adhering to regulatory authorities is extremely important when concerning food along with keeping good relations with the suppliers of the food statistics if the tool is to be kept up to date.

Internal

Project Team

- Project Manager
- Programmers/software developers
- Data analysts
- Cyber Security Analyst
- Business Analyst

Marketing Team

Customer Support Team

Board of Directors

External

Customers

- Health Conscious individuals
- Athletes & Coaches

- People with dietary restrictions
- Researchers/Universities

Investors

Regulatory Agencies

Suppliers and Collaborators

1.3 Project Scope

Inside the Project Scope –

These features will be offered by the system and will offer the user a thorough analysis of each of the foods nutritional data

• Allow users to search foods by name and see key nutritional statistics

• Allow these stats to be broken down and displayed as bar and pie charts for further analysis of the data.

• Enable users to select a nutritional element, input a maximum and minimum value and display all foods which fall within this range.

• Enable users to filter foods by nutritional content levels, falling within a range of high medium or low.

• Enable users to compare two foods side by side using both the key nutritional statistics and the bar and pie chart page to do so. !Update: Comparison of foods is only available as a Bar chart as it shows a far better comparison of the two foods. The comparison with the pir chart was considered to be inferior and unnecessary.!

• Enable users to sort foods from highest to lowest and any nutritional statistic. !Update: Removed due to being outside the needed project scope. Was originally considered as the additional feature but was deemed not as beneficial.!

• Have easily laid out code dictionaries in which foods can be added by collaborators in the future !Update: Considered Too Difficult to implement and outside project Scope.!

Outside the Project Scope –

These features will not offered within the project as they are either too complex or deemed not necessary to the projects success.

• Allow users to import or update food details/statistics • Allow users access directly to the data used

• Allow users to compare more than 2 foods at one time

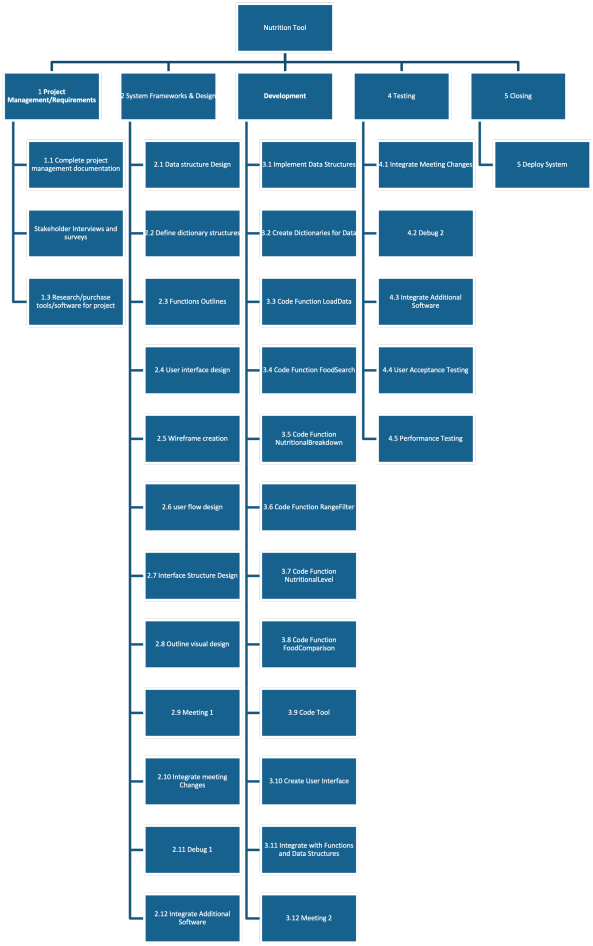
• Allow users to search for foods which meet more than one nutritional criteria

• Allow users have secondary filters when sorting from highest to lowest.

• Allow users to filter by ranges not specified above in the project scope

2. Work Breakdown Structure

Within the Work breakdown structure both a hierarchy and a table have been included to show further understanding of what tasks need to be completed in order to move forward in the project. Core themes of the project are broken down into deliverables, required to be completed before moving to the next theme. These themes are broken down into five sections; Project Managements/Requirements, System Framework and Design, Development, testing and closing.



Work_Breakdown_Structure.png

The table below further outlines the workbreakdown structure and what task can be completed in tandem and the dependencies for each one.

Work Breakdown Structure Food Nutrition Tool			
Task ID	**Task Description**	**Duration (Days)**	**Predecessor**
	Project Management/Requirements		
1.1	Complete project management documentation	4	
1.2	Stakeholder Interviews and surveys	4	
1.3	Research/purchase tools/software for project	2	1.1, 1.2
	System Frameworks & Design		
2.1	Data structure Design	1	1.3
2.2	Define dictionary structures	2	1.3
2.3	Functions Outlines	1	1.3
2.4	User interface design	2	1.3
2.5	Wireframe creation	2	2.4
2.6	user flow design	1	2.4, 2.5
2.7	Interface Structure Design	2	2.6
2.8	Outline visual design	2	2.7
2.9	Meeting 1	1	2.8
2.10	Integrate meeting Changes	1	2.9
2.11	Debug 1	4	2.9, 2.10
2.12	Integrate Additional Software	4	2.9, 2.10
	Development		
3.1	Implement Data Structures	1	2.12
3.2	Create Dictionaies for data	2	3.1
3.3	Implement/code Function LoadData	2	3.1, 3.2
3.4	Implement/code Function FoodSearch	3	3.1, 3.2, 3.3
3.5	Implement/code Function NutritionalBreakdown	3	3.1, 3.2, 3.3
3.6	Implement/code Function RangeFilter	3	3.1, 3.2, 3.3
3.7	Implement/code Function NutritionalLevel	3	3.1, 3.2, 3.3
3.8	Implement/code Function FoodComparison	3	3.1, 3.2, 3.3
3.9	Code Tool	7	3.4, 3.5, 3.6, 3.7, 3.8
3.10	Create User Interface	6	3.9
3.11	Integrate with Functions and Data Structures	3	3.1
3.12	Meeting 2	1	3.11
	Testing		
4.1	Integrate meeting Changes	2	3.11
4.2	Debug 2	5	4.1
4.3	Integrate Additional Software	1	4.2
4.4	User Acceptance Testing	3	4.3
4.5	Performance testing	3	4.4
5	Deploy System	1	4.5

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3. Activity Definition Estimation

Within the project different members of the team will be working on different parts at various times. All of the activities have been broken down and further explained within the description. A duration for each activity has also been defined to give the project an estimated overall completion time.

Activity Definition Estimation				
Activity Number	Activity Name	Brief Description	Duration	Responsible Team Members
1.1	Complete project management documentation	Complete all necessary documentation and planning after being approached by business about project. This includes the Project Plan and beginning the System Design Document	4	Project Manager
1.2	Stakeholder Interviews and surveys	These interviews and surveys will be used to gain further insight into the product that the business is looking for. Changes can be made to the documentation after the results to better reflect the users need.	4	Project Manager, Business Analyst
1.3	Research/purchase tools/software for project	Purchase any hardware needed for the completion of the project. Also make sure any licences for software needed have been purchased. Ensure that the work space is ready for the development of the project to fully commence.	2	Project Manager
2.1	Data structure Design	Brainstorm and create a high level plan/design of how the Food data will be stored for the completed product.	1	Head Programmer, Data Analysts, Software Developers
2.2	Define dictionary structures	Go into more detail on how each of the subcategories and data points will be defined and stored. Further elaborate on how each set of data will be stored and the hierachry of said data.	2	Head Programmer, Data Analysts
2.3	Functions Outlines	Outline and define how each function will work and how the end user will interact with these functions. List the input values along with the intended output values that a user can expect to receive.	1	Head Programmer, Project Manager
2.4	User interface design	Create a high level design for how the tool will look to the end user. This will have low levels of detail and be focused mainly around building a design which is intuitive, understandable and easy to use whilst also looking visually appealing.	2	Head Programmer, Business Analyst, Project Manager
2.5	Wireframe creation	Using the interface design the team will create wireframes for how each individual page will look. This will go into more depth on how the features will be implemented into the overall design. It is important to always refer back to the interface design to ensure the wireframes meet all the functional and visual requirements outlined previously.	2	Project Manager, Head Programmer
2.6	user flow design	An in depth flow chart of how a user will use the product and how navigation/tasks will be performed using the tool. This is exremely important to showcase how a user will interact with the product and make sure that the design and navigation is intuitive and logical.	1	Head Programmer, Business Analyst
2.7	Interface Structure Design	A design phase to insure that all previous steps in the design phase are coherent and working as intended with each other. This step will take advantage of having all other design areas completed and making sure that what has been created has an overarching purpose behind it.	2	Head Programmer
2.8	Outline visual design	taking heavily from task 5, task 9 will focus on the details of the design of the product ensuring that once everything has been deemed suitable together a consistent theme and style can be property applied to the product.	2	Head Programmer, Business Analyst, Project Manager
2.9	Meeting 1	First team meeting at the end of the design phase to ensure the team are all in agreement and happy to move forward to the next phase of the project. This is where issues or questions can be raised and addressed before they become too far gone.	1	All
2.10	Integrate meeting Changes	Any changes agreed upon in the meeting will be implemented here	1	All
2.11	Debug 1	Not directly debugging as very little code will have been written at this point however it is a time where everything up to this point, including the meeting changes can be reviewed and ensure the project is on track.	4	All
2.12	Integrate Additional Software	If any issues arise during the debugging phase extra days have been allocated to ensure the project is ready to move into full development. These days can also be seen as a fail safe in case any of the above tasks take longer than orignally estimated.	4	All
3.1	Implement Data Structures	The first phase of the development phase is to implement how the data will be stored for the product.	1	Head Programmer, Data Analysts, Software Developers
3.2	Create Dictionaies for data	This phase will create the specific data structures and ensure that all structures outlined within the documentation and planning have been implemented.	2	Head Programmer, Data Analysts, Software Developers
3.3	Implement/code Function LoadData	Implement the function that will read the Nutritional Food Database into the software, and store the data within as object/dictionary.	2	Head Programmer, Software Developers
3.4	Implement/code Function FoodSearch	Implement the function that is responsible for sorting through the dictionary where the food items are stored, and checking whether a food item name matches what the user has input into the food search text input.	3	Head Programmer, Software Developers
3.5	Implement/code Function NutritionalBreakdown	Implement the function thats purpose is to draw pie charts and grapha for the food item the user has selected. This will happen when the user presses a button while having a food item selected, or when they select the pie chart or graph button in the graph area itself.	3	Head Programmer, Software Developers
3.6	Implement/code Function RangeFilter	Implement the fucntion that will be used when the Range Filter option has been enabled. It will take several user inputs to act as a filter when displaying food item search results.	3	Head Programmer, Software Developers
3.7	Implement/code Function NutritionalLevel	Implement the function that will allow a user to select a type of nutritional value from a dropdown list(fat, protein, carbohydrates, sugar or nutritional density), and then select one of three options, low, medium or high, and have the search results filtered by these decisions.	3	Head Programmer, Software Developers
3.8	Implement/code Function FoodComparison	Implement the function that will allow the user to directly compare two foods and their nutritional values. This tool will be extremely helpful for those trying to compare two similar items or who are trying to find healthier alternitives to foods they currently eat.	3	Head Programmer, Software Developers
3.9	Code Tool	Implementing the designs that have been previously outlined and creating what the end user will ultimately see.	7	Head Programmer, Software Developers, Business Analyst
3.10	Create User Interface	Closely tied with task 20 ensuring that the interface is the same as the design template and ensuring that navigation and the users flow through the product are all as intended.	6	Head Programmer, Software Developers, Business Analyst
3.11	Integrate with Functions and Data Structures	combine the functions and data structures with the UI that has been created and ensure that the products parts are all working together as one. This step should closely resemble the final product.	3	Head Programmer, Software Developers
3.12	Meeting 2	Now that a majority of development is complete the team will now reconvene and ensure that the tool is working/looking as intended.	1	All
4.1	Integrate meeting Changes	Any changes agreed upon in the meeting will be implemented here	2	All
4.2	Debug 2	This time will beused to ensure that the whole project is working before sending it to testing. Making sure that any issues that may arise in testing can be dealt with as early as possible.	5	All
4.3	Integrate Additional Software	Integrate any additional changes that have been agreed upon in the meeting. Similar to the design phase this section is meant to give some leeway if certain tasks take longer than anticipated.	1	All
4.4	User Acceptance Testing	This phase involves verifying that the system meets the business requirements and is ready for use by the end-users. Test cases are executed based on real-world scenarios to ensure that the system performs as expected, with feedback collected to identify any issues that need to be addressed before deployment	3	All
4.5	Performance testing	This task focuses on assessing the system's responsiveness, stability, and scalability under various conditions. By simulating different loads and stress levels, performance testing ensures that the system can handle peak usage without degradation, maintaing optimal performance for users	3	All
5	Deploy System	Deployment involves moving the system from a development or testing environment to a live, production environment. This task includes setting up the necessary infrastructure, configuring the system for production, and ensuring that all components are properly integrated and functioning, making the system available for end-users.	1	All

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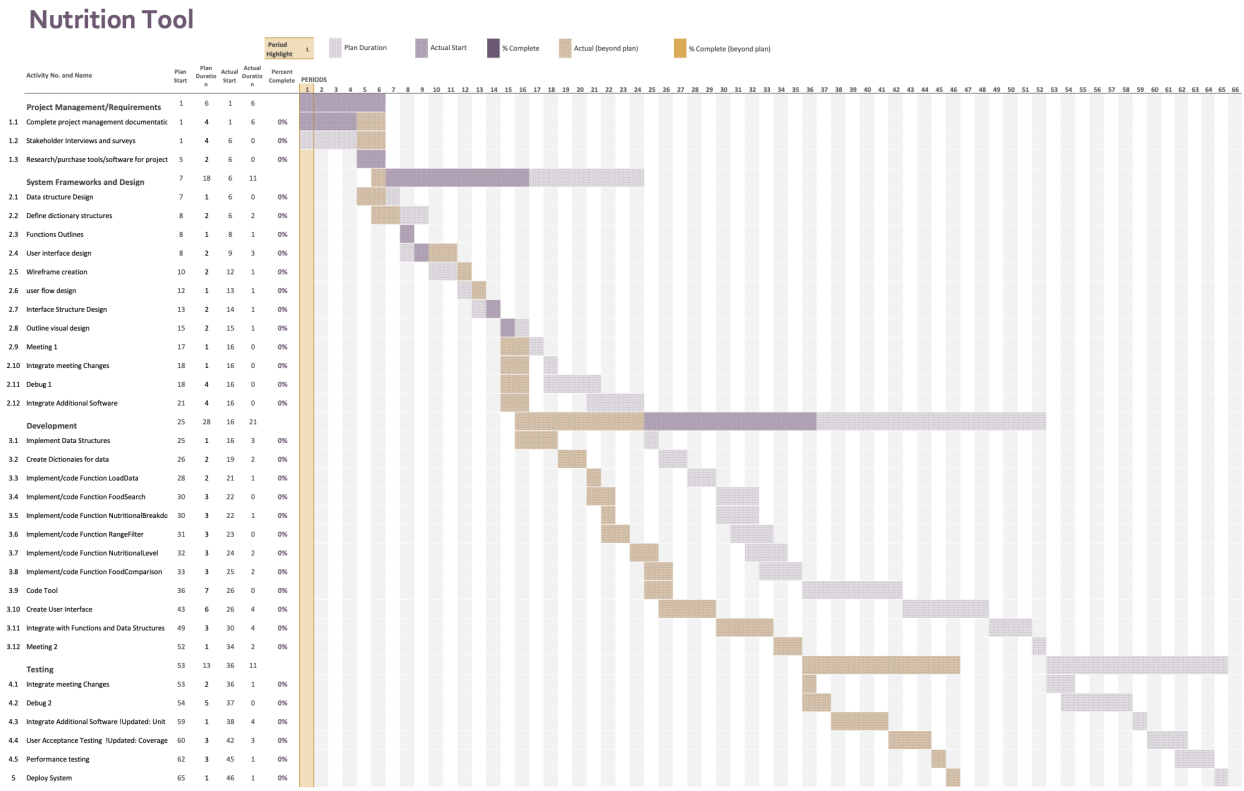
4. Gantt Chart

The Gantt chart for this project provides a visual timeline of key tasks, illustrating their start and end dates as well as their dependencies. It serves as a comprehensive schedule, allowing for clear tracking of project progress and ensuring that milestones are met on time. By outlining the duration and overlap of each activity, the chart aids in effective project management and resource allocation.

!Update: The Gantt Chart over estimated the amount of time it would take us to complete many activities. In particular at the end of the design phase which was much smoother than anticipated. Some areas towards the end were update to better reflect the testing process.



	Activity No. and Name	Plan Start	Plan Duration	Actual Start	Actual Duration	Percent Complete
	Project Management/Requirements	1	6	1	6	
1.1	Complete project management documentatic	1	4	1	6	0%
1.2	Stakeholder Interviews and surveys	1	4	6	0	0%
1.3	Research/purchase tools/software for project	5	2	6	0	0%
	System Frameworks and Design	7	18	6	11	
2.1	Data structure Design	7	1	6	0	0%
2.2	Define dictionary structures	8	2	6	2	0%
2.3	Functions Outlines	8	1	8	1	0%
2.4	User interface design	8	2	9	3	0%
2.5	Wireframe creation	10	2	12	1	0%
2.6	user flow design	12	1	13	1	0%
2.7	Interface Structure Design	13	2	14	1	0%
2.8	Outline visual design	15	2	15	1	0%
2.9	Meeting 1	17	1	16	0	0%
2.10	Integrate meeting Changes	18	1	16	0	0%
2.11	Debug 1	18	4	16	0	0%
2.12	Integrate Additional Software	21	4	16	0	0%
	Development	25	28	16	21	
3.1	Implement Data Structures	25	1	16	3	0%
3.2	Create Dictionaies for data	26	2	19	2	0%
3.3	Implement/code Function LoadData	28	2	21	1	0%
3.4	Implement/code Function FoodSearch	30	3	22	0	0%
3.5	Implement/code Function NutritionalBreakdo	30	3	22	1	0%
3.6	Implement/code Function RangeFilter	31	3	23	0	0%
3.7	Implement/code Function NutritionalLevel	32	3	24	2	0%
3.8	Implement/code Function FoodComparison	33	3	25	2	0%
3.9	Code Tool	36	7	26	0	0%
3.10	Create User Interface	43	6	26	4	0%
3.11	Integrate with Functions and Data Structures	49	3	30	4	0%
3.12	Meeting 2	52	1	34	2	0%
	Testing	53	13	36	11	
4.1	Integrate meeting Changes	53	2	36	1	0%
4.2	Debug 2	54	5	37	0	0%
4.3	Integrate Additional Software !Updated: Unit	59	1	38	4	0%
4.4	User Acceptance Testing !Updated: Coverage	60	3	42	3	0%
4.5	Performance testing	62	3	45	1	0%
5	Deploy System	65	1	46	1	0%

Gantt_Chart_A1.png